Introduction to Data Science

Semester Project — Full Guidelines

Covers the Complete Data Science Life Cycle | R Markdown Implementation | Group + Individual Assessment

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Project Life Cycle — Phase Overview

Phase 1 — Midterm Upstream stages | Deadline: 18 July 2026	**Phase 2 — Final** Downstream stages | Deadline: 29 August 2026
Section 1 — Problem framing Domain context, research questions, KPIs	**Section 5 — PCA** Dimensionality reduction, scree plot, biplot
Section 2 — Data acquisition Load from URL, inspect, document provenance	**Section 6 — Modelling** Baseline + improved model, cross-validation
Section 3 — Exploratory data analysis Distributions, skewness, correlations, visuals	**Section 7 — Evaluation** Metrics, comparison table, visualisation
Section 4 — Preprocessing & features Cleaning, engineering, selection, transform	**Section 8 — Communication & reflection** Executive summary, visuals, limitations
Deliverable: SectionGroupNoMid.Rmd + .html	**Deliverable: SectionGroupNoFinal.Rmd + .html**

1. PROJECT OVERVIEW

This semester project is designed to give students hands-on experience with every stage of the data science life cycle. The project runs across two phases — Midterm and Final — using a single, continuously evolving R Markdown document. Each phase builds directly on the previous one, culminating in a complete, reproducible data science report.

Students work in groups. Each group submits one shared .Rmd file and one rendered .html file per phase. The group mark (30 marks per phase) is awarded for the quality of the submitted document. The individual mark (20 marks per phase) is awarded separately through a viva conducted for each student individually, assessing their understanding of the project implementation and their performance on related lab tasks.

	Phase 1 — Midterm	Phase 2 — Final
Deadline	18 July 2026 (see note below)	29 August 2026
Focus	Upstream stages (data to features)	Downstream stages (models to insights)
Group marks	30	30
Individual marks	20	20
Total	50	50
Deliverables	One .Rmd + one .html per group	One .Rmd (extended) + one .html per group

2. DATASET SELECTION REQUIREMENTS

Each group selects one real-world dataset from an approved public repository. The dataset must meet all of the following criteria:

- Minimum 500 rows and 8 variables
- A clearly definable target variable suitable for regression or classification
- A mix of numeric and categorical features
- Contains some messiness: missing values, outliers, or inconsistent encoding
- Publicly accessible via a direct URL (GitHub raw link, Google Drive, or an open repository such as UCI ML Repository, Kaggle, data.gov, WHO, or World Bank)

> *Important: Students do NOT submit the dataset file. The data must be loaded directly inside the R Markdown document from a public URL using read.csv(), readr::read_csv(), or an equivalent function. The code must run without any modification on any machine.*

Each group submits a one-paragraph dataset justification for approval before work begins.

3. TECHNICAL & SUBMISSION REQUIREMENTS

R Markdown Configuration

The YAML front matter of the .Rmd file must include:

- output: html_document with toc: true, toc_float: true, and code_folding: hide
- A meaningful title, author (all group member names), and date field
- set.seed() declared immediately after the YAML block for full reproducibility

Package Installation

All required packages must be installed dynamically inside the document using the following pattern, so the document runs on any machine without manual setup:

```
if (!require("tidyverse")) install.packages("tidyverse")
if (!require("skimr"))     install.packages("skimr")
if (!require("caret"))     install.packages("caret")
library(tidyverse)
library(skimr)
library(caret)
```

Submission

- Each group submits one .Rmd source file and one rendered .html file — not one per member
- The .Rmd must knit successfully without any modification or manual intervention
- File naming convention: SectionGroupNoMid.Rmd / SectionGroupNoMid.html for Phase 1; SectionGroupNoFinal.Rmd / SectionGroupNoFinal.html for Phase 2
- Example: J01Mid.Rmd and J01Mid.html for Phase 1; J01Final.Rmd and J01Final.html for Phase 2
- One member of the group is responsible for submitting both files to the MS Teams inbox
- Individual marks are NOT based on the document — they are assessed through a viva for each student
- Late submissions: 10% deduction per day unless an extension is granted in advance

> *The code must run from start to finish without modification on any standard R installation. If it fails to knit during evaluation, the submission will be penalised. Test your document on a fresh R session before submitting.*

> *For any clarifications regarding the project requirements, technical issues, or expectations, students are welcome to attend consulting hours.*

4. PHASE 1 — MIDTERM SUBMISSION (DEADLINE: 18 JULY 2026)

Phase 1 covers the upstream stages of the data science life cycle. Students build and submit a self-contained R Markdown report containing the following sections:

Section 1 — Problem Framing

- Describe the domain context and dataset background
- State 2–3 specific, testable analytical questions (not just "predict X")
- Define success criteria and identify the relevant evaluation metrics
- Justify why this dataset is appropriate for the stated questions

Section 2 — Data Acquisition & Initial Inspection

- Load data from a public URL using read_csv() or equivalent — no local file paths
- Report dimensions, variable types, and a printed head() of the dataset
- Produce a full summary statistics table using skimr::skim()
- Document data provenance: source, collection method, and known limitations

Section 3 — Exploratory Data Analysis (EDA)

- Univariate analysis: distribution plots for all key variables (histograms, bar charts via ggplot2); each plot must be followed by a short observation documenting the key pattern or distribution shape
- Skewness analysis: compute skewness for each numeric variable (e.g. using moments::skewness() or e1071::skewness()); document for each variable whether transformation is warranted and why
- Bivariate analysis: scatter plots, box plots, and grouped summaries linking predictors to the target variable; each output must be followed by a short observation on the relationship observed
- Correlation analysis: correlation matrix or heatmap for numeric variables; document which pairs show strong positive or negative correlation and any multicollinearity concerns
- At least one non-obvious or interesting finding must be highlighted with a supporting plot and a short written observation

Section 4 — Preprocessing & Feature Engineering

- Missing value treatment: imputation or removal with explicit justification for each decision
- Outlier detection and treatment: document the strategy (IQR, Z-score, or domain reasoning)
- Categorical encoding: dummy/one-hot encoding or ordinal encoding as appropriate
- Numeric scaling/normalisation: standardisation or min-max scaling where justified
- Feature engineering: create at least two new meaningful features derived from existing variables (e.g. ratios, bins, interaction terms, date decomposition, log transforms); justify each
- Feature selection: apply at least one feature selection technique (e.g. correlation filter, variance threshold, or recursive feature elimination) and document which features are retained and why
- Save the cleaned, engineered, and selected feature set as a data frame for use in Phase 2

> *Feature engineering and feature selection are assessed separately. Simply dropping columns does not constitute feature selection — a principled, documented method is required.*

5. PHASE 2 — FINAL SUBMISSION (DEADLINE: 29 AUGUST 2026)

Phase 2 extends the same .Rmd file. Students append the following sections to the document submitted at midterm, producing one cumulative report.

Section 5 — Dimensionality Reduction with PCA

- Apply Principal Component Analysis (PCA) to the numeric features of the preprocessed dataset using prcomp() or princomp()
- Plot the scree plot and cumulative explained variance; document how many components are selected and what percentage of variance they explain
- Produce a biplot or loading plot; document which original features contribute most to the first two components
- Document whether PCA improves or hinders model interpretability for the specific dataset, with reasoning
- Students may choose to train models on PCA-transformed features or on the original feature set; both choices must be justified

Section 6 — Modelling

- Split the dataset into training and test sets (e.g. 80/20 split) using rsample or caret
- Fit at least two models: one baseline (linear or logistic regression) and one improved model (decision tree, random forest, k-NN, or similar)
- Apply k-fold cross-validation (k = 5 or 10) for model selection and hyperparameter tuning
- Document all modelling decisions: algorithm choice, hyperparameter values, and rationale

Section 7 — Evaluation

- Report appropriate metrics — RMSE and R^2 for regression; accuracy, precision, recall, F1, and ROC-AUC for classification
- Compare all models in a formatted kable table; follow the table with a short observation identifying the best-performing model and possible reasons
- Visualise predictions vs. actuals (regression) or the confusion matrix (classification); document key observations from the visualisation
- Interpret results in domain terms, not just numerical summaries

Section 8 — Communication

- Write an executive summary (200–300 words) suitable for a non-technical stakeholder
- Produce at least one polished, publication-quality ggplot2 figure that directly answers one of the original research questions from Section 1
- Each code chunk must be followed by a short observation or interpretation of the output where applicable; avoid isolated code blocks with no context

Section 9 — Reflection

- Discuss at least two specific limitations of the analysis (data quality, model assumptions, scope)
- Comment on ethical considerations: potential biases in the data, fairness of predictions, and privacy implications
- Propose two concrete, actionable directions for future work

6. WEEKLY MILESTONES & SCHEDULE

The table below shows the recommended weekly targets. Milestones are advisory and not separately graded, but brief progress checks may be conducted.

Week	Dates	Phase	Milestone Target
Week 1	26 Jun – 2 Jul	Phase 1	Dataset approved; Section 1 (Problem Framing) drafted; data loads from URL
Week 2	3 Jul – 9 Jul	Phase 1	Sections 2 & 3 complete: data inspection, EDA with skewness analysis, correlation heatmap
Week 3	10 Jul – 17 Jul	Phase 1	Section 4 complete: preprocessing, feature engineering, feature selection; final review and clean knit
★Week 3	20 Jul	Phase 1	MIDTERM DEADLINE — .Rmd + .html submitted. Note: midterm exams begin 19 July — aim to submit by 18 July
Week 4	19 Jul – 25 Jul	Phase 2	Section 5: PCA implemented, scree plot, biplot, component interpretation
Week 4	19 Jul – 25 Jul	Phase 2	Section 6: baseline and improved models trained with cross-validation
Week 5	26 Jul – 1 Aug	Phase 2	Section 7: evaluation metrics, model comparison table, prediction visualisation
Week 6	2 Aug – 8 Aug	Phase 2	Section 8: executive summary, polished ggplot2 figure, narrative review
Week 7	9 Aug – 28 Aug	Phase 2	Section 9: reflection; full document review, knit test on a clean session
★Week 7	29 Aug	Phase 2	FINAL DEADLINE — complete cumulative .Rmd + .html submitted

> *Weeks marked ★are hard deadlines. Midterm exams begin on 19 July — students are strongly advised to complete and submit Phase 1 before exam week begins. It is recommended to complete a clean knit of the full document at least two days before each deadline.*

7. MARKING RUBRIC

Phase 1 — Midterm (Total: 50 marks)

Group mark (30 marks):

Component	Marks	Key Criteria
Problem framing & research questions	5	Clear domain context, specific & testable questions, defined success metrics
Data acquisition & provenance	5	Loads from URL, correct inspection, documented limitations
Exploratory Data Analysis (incl. skewness)	10	Complete univariate/bivariate analysis, skewness computed & interpreted; informative visuals
Preprocessing, feature engineering & selection	10	All cleaning decisions justified; ≥2 engineered features; principled feature selection method

Individual mark (20 marks) — awarded via viva:

Component	Marks	Key Criteria
Understanding of project implementation	12	Student can explain data choices, preprocessing steps, EDA findings, and code logic when questioned
Lab task performance	8	Performance on related lab tasks assigned during the course

> *The individual viva is conducted separately for each student. Students must be able to explain any part of the group submission. Not being able to explain the group's code or decisions will result in a lower individual mark regardless of document quality.*

Phase 2 — Final (Total: 50 marks)

Group mark (30 marks):

Component	Marks	Key Criteria
PCA implementation & interpretation	8	Correct prcomp(), scree plot, biplot, explained variance, justified component choice
Modelling & cross-validation	10	Baseline + improved model, k-fold CV, hyperparameter documentation
Evaluation rigor & comparison	7	Correct metrics, model comparison table, prediction visualisation, domain interpretation
Communication & polished visuals	5	Executive summary (200–300 words), publication-quality ggplot2 figure

Individual mark (20 marks) — awarded via viva:

Component	Marks	Key Criteria
Understanding of project implementation	12	Student can explain modelling choices, PCA, evaluation results, and code logic when questioned
Lab task performance	8	Performance on related lab tasks assigned during the course

> *The individual viva is conducted separately for each student. Students must be able to explain any part of the group submission. Not being able to explain the group's code or decisions will result in a lower individual mark regardless of document quality.*

8. ACADEMIC INTEGRITY & USE OF AI TOOLS

All code and written analysis must be the original work of the group. The following policies apply:

- Direct copying of code from online sources without attribution is prohibited.
- AI-generated code or text may be used as a learning aid but must be disclosed in a brief note within the document. Students are responsible for understanding and being able to explain every line of code.
- All group members are expected to contribute equitably. Individual accountability is assessed through the viva — every student must be able to explain the full project, not just their own portion.
- Plagiarism between groups will result in zero marks for all parties involved.

9. RECOMMENDED R PACKAGES

Package	Purpose
tidyverse	Core data manipulation and ggplot2 visualisation
skimr	Compact summary statistics for EDA
moments / e1071	Skewness and kurtosis computation
knitr + kableExtra	Formatted tables in HTML output
caret / tidymodels	Model training, cross-validation, and evaluation
rsample	Train/test splitting and resampling
ggcorrplot	Correlation matrix heatmaps

factoextra	PCA visualisation (scree plot, biplot)
randomForest / rpart	Tree-based improved models
pROC	ROC curve and AUC computation

> *All packages must be installed dynamically using if (!require(...)) install.packages(...) — never assume they are pre-installed on the evaluator's machine.*